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RESEARCH ARTICLE

Security Enhanced CLS and CL-AS Scheme Without Pairings for VANETs

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ABSTRACT While vehicle ad-hoc networks (VANETs) have many advantages, they also present privacy and security concerns. Certificate management issues have been seen in traditional public key infrastructure-based privacy-preserving authentication schemes, while key escrow problem exists in identity-based privacy-preserving authentication techniques. Also, existing cryptographic techniques rely heavily on assumptions about tamper-proof equipment to ensure their security. A proposal has been made for a certificateless aggregate signature system for VANETs addressing these issues and was proved provably unforgeable against collusion attack. However, we found that the proposed technique was insecure and could not withstand collusion assault. Therefore, this paper presents an improved and secure certificateless aggregate signature technique for VANETs. We also illustrate the security & performance evaluation of our presented technique, and based upon the hardness assumption of the elliptic curve discrete logarithm problem, we have shown that the technique is safe against existential forgery on adaptive chosen message attack in the random oracle model. Also, the presented technique has better efficiency compared to some recent existing authentication techniques.

INDEX TERMS Aggregate signature, certificateless public key cryptography, elliptic curve cryptosystem, vehicular ad hoc networks.

I. INTRODUCTION

As wireless communication technology advances, automobiles outfitted with wireless gadgets can now communicate with one another. Vehicular ad hoc networks (VANETs) have currently gained trend as well as attention. VANETs give an innovative approach to utilize automobiles [4], [25], [31] to enhance traffic security, reduce environmental pollution, relieve traffic jams, and provide comfort [29]. Every vehicle functions as a router, communicating via embedded wireless devices called on-board units (OBUs). A common VANET structure should have vehicles deployed with on-board units (OBUs), Road side units (RSUs), and a trusted authority (TA). There are 3 types of communications available: vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I) & infrastructure-to-infrastructure (I2I). V2I and I2I communications use encrypted channels, but V2V communication employs an open dedicated short-range communication (DSRC) protocol. In every 100 to 300 ms, OBUs broadcast traffic-related messages like the car's position, present time, vehicle speed and direction, and traffic situation [9], [14], [29].

VANETs offer numerous benefits but it also faces certain challenges among which security and privacy are the main

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concern. VANETs perform various functions like traffic management, providing safety-related information, helping to avoid traffic accidents and minimising traffic jams. If the VANET network is not secure, it can provide wrong information, resulting in various problems created on the road. This is why a VANET network should be secure, and therefore, security is one of the important considerations regarding VANETs.

The other key issue in VANET is privacy. A digital signature is a broad cryptographic technique that brings confidentiality, integrity, & non-repudiation. With the adoption of digital signature, VANETs provide message authentication & integrity, privacy, & non-repudiation [10]. One of the important security issues in VANETs is authenticity. Using traditional public key cryptography (PKC) [17], [26], [33], identity-based cryptography [40] and group signature-based authentication [23], [27], several authentication mechanisms have been proposed. In PKC, a huge quantity of public key certificates are to be maintained for which a certificate authority is required due to which techniques undergo the certificate management problem. Also, group signature-based schemes have higher communication costs, which makes them impractical to use. The identity (ID) based techniques address the key escrow problem and are appropriate for private networks only.

Al-Riyami and Paterson [1] have proposed the certificateless public key cryptography that settles the key escrow problem in ID-based schemes in which the third party, known as the key generation center, creates the user's partial private key while the user creates his private key. Due to which malicious KGC cannot access the user's private key. In 2003, the term aggregate signature was brought into concern by Boneh et al. [5] in which n signatures on n distinct messages are aggregated into one signature which results in high efficiency. The certificateless aggregate signature(CLAS) method is an effective digital signature scheme that combines the advantages of aggregate signature & certificateless signature (CLS) by addressing the certificate issue in regular PKC & the key escrow problem in ID-PKC. It reduces calculation & transmission expenses by utilizing the efficiency advantage of aggregate signatures.

In VANETs some CLAS schemes have been proposed [9], [15], [22]. However, some schemes have higher communication and computation overhead, while some are insecure and cannot be employed for practical applications. The main research challenge is to produce an efficient and secure certificateless signature scheme which also reduces the communication and computation overhead. In [9], an efficient CLAS scheme without pairing has been proposed. However, we prove that their technique is not secure against forgery attacks & we give a fresh & secure CLS and CLAS technique for VANETs.

Our contribution: Based on elliptic curve cryptography, we present a fresh CLS technique for VANETs. Following this CLS technique, we present a CL-AS strategy for V2I communications in VANETs. To be precise, the primary contributions of this study are as follows.

- We initially discover some security issues in Cui et al.'s CLS & CLAS technique.
- We then present a new and secure CLS & CLAS technique for VANET.
- Further, we gave the security analysis of the suggested technique & show that the technique presented is existentially unforgeable against forgery attack under random oracle model (ROM) [3].
- Finally, we analyze the computation expense of our suggested system and give a brief analogy of the presented technique with some related techniques for VANETs.

The arrangement of the remaining paper is as follows: Section II presents some related work on the certificateless aggregate signature scheme. Some preliminaries and structure of the aggregate signature technique are given in Section III. We reviewed the existing CLS & CLAS technique by Cui et al. in section IV & section V gives the forgery attack on Cui et al.'s signature technique. In section VI, we presented our new CLS and CLAS technique for V2I communication in VANET. Section VII presents the detailed security proof of the presented technique. In section VIII security analysis & performance evaluation of the presented technique are given, and lastly, we wind up section IX with the conclusion.

II. RELATED WORK

Digital signature [18] is a significant component of public key cryptography, which enables anybody to confirm the validity of a signature by the use of the public key of the sender. Digital signatures can be used successfully in both one-to-many (multicast) & one-to-one (unicast) applications because of this feature. Applications for multicasting might enable the root node to gather information from leaf nodes, resulting in a pattern of many-to-one communication [28], [43].

Boneh et al. [5] originally presented the idea of an aggregate signature in 2003. An aggregate signature is created when an aggregation algorithm combines the signatures over n different messages, which are signed by n users, totalling n. The verifier needs to verify that the aggregate signature should be from the designated n users. Aggregate signatures can be useful in situations when bandwidth and storage are limited since they lower the cost of verification while also potentially reducing the sent signature's length.

Numerous further aggregate signature techniques have been presented after the proposal of Boneh et al. Nevertheless, there have been difficulties in putting the notion of signature aggregation into practice. Typically, each user has its own public key & private key. Certificate authorities (CAs), which are the third parties, make up the public key infrastructure (PKI) and provide the public key. A digital certificate that is casted by the CA provides the connection between a user & his public key. Managing certificates demands a significant quantity of computation as well

as storage power since the recipient of communication must obtain this certificate to confirm its ownership and authenticity.

Shamir et al. [32] suggested identity-based public key cryptography (ID-PKC) in 1984 for the certificate management processes of PKI. This technique generates users' public keys based on their mobile number or email address. To get a user's private key, though, they need a Private Key Generator (PKG), which is a reliable third party. The master secret key of the PKG & the user's publicly known identity are used to generate his private key. As a result, there is still an issue with the key escrow in ID-PKC, meaning that every user must have complete faith in the PKG. This may be an overly strong and unrealistic assumption in many real-world situations.

Al-Riyami and Paterson [1] gave a novel paradigm known as certificateless public key cryptography (CL-PKC) to resolve the key escrow issue with ID-PKC. Although a user must use a third party, known as a Key Generation Center (KGC), in order to create his private key under CL-PKC, there is no public key certificate in certificateless cryptography [32], [41]. Because the KGC merely creates a partial private key for each user, this prevents the KGC from having accession to the entire private key of that user. The partial private key they obtained and a user's secret value make up the complete private key in CL-PKC. This makes it possible for CL-PKC to get around the difficult certificate administration issue in conventional PKI systems & the key escrow issue in ID-PKC systems.

The aforementioned benefits of CL-PKC have greatly stimulated additional study. Yum and Lee [39] then offered a generic construction of the certificateless signature (CLS) technique, which was first suggested by Al-Riyami & Paterson [1]. Hu et al. [16], however, demonstrated the shortcomings of Yum and Lee's [39] structure and suggested a solution for the conventional model. In their report that outlines their CLS plan, after Al-Riyami and Paterson [1] failed to produce any legal evidence, Huang et al. [15] discovered a security flaw in the plan, offered a safe workaround, and the initial security model is created by them. Zhang et al. [45] developed a more effective CLS technique in 2006 by enhancing the security model of the previous CLS schemes. Researchers are trying to work on safe and effective certificateless aggregate signing methods up to this point [11], [43], [44]. Nonetheless, the disadvantage of most of the suggested methods is that they necessitate intricate pairing calculations in the verification procedure, which escalates proportionally with the number of users [6], [11], [44]. This goes against the fundamental objective of aggregate signatures, which is to reduce the amount of work involved in the verification process.

A CLAS system was suggested by Xiong et al. [37] in 2013. It is presumed on certificateless public key cryptography and requires a fixed quantity of pairing computations for validation, regardless of the total number of aggregated signatures. Regretfully, He et al.'s [13] plan & Cheng et al.'s [8] system demonstrate the insecurity of their scheme. Moreover, it has been discovered that the impersonation assault can be used to undermine the strategy of Xiong et al. [37]. An opponent could, for instance, counterfeit a legitimate signature on any communication.

Studies over VANETs security and solutions have also been made. Anwar et al. [46] proposed a survey of VANET technologies in which they analysed and compared the wireless standards using various parameters such as bandwidth, ease of use, upfront cost, maintenance, accessibility, signal coverage, signal interference and security. Further, in 2017, Hasrouny et al. [47] proposed a survey of VANET security and challenges where they gave the details of the recent security architectures and the well-known security standards protocols in VANET infrastructure. Several more studies have also been made on enhancing VANETs security.

A novel CLAS scheme was suggested by Liu et al. [24] in 2014, and they showed the security of their scheme in the ROM. Liu et al.'s [24] suggested his scheme, but it cannot provide unforgeability. Chen et al. [7] introduced a novel certificateless signature technique that was later modified to become a secure certificateless aggregate signature scheme. Although they said that their system is resistant to the attacks mentioned above, it has since been demonstrated in [42] that this CLS technique is likewise unsecured and open to a more potent attack, wherein anybody can counterfeit a signature without a certificate on any message.

A CL-AS method with bilinear pairing was presented by Kumar et al. [22]. In 2018, Cui et al. [9] presented a CLAS technique without bilinear pairing & claimed that the presented technique is secure against forgery attack in the ROM. Kamil et al. [20] observed that the method was insecure and they gave the improved method and proved that their scheme is safe in the ROM. A CLAS technique against a fully chosen key attack has been suggested by Wu et al. [36]. A provably secure & efficient certificateless short signature (CLSS)-based authentication technique with conditional privacy-preserving (CLSS-CPPA) has been presented by Ali et al. [2] for V2V communication. Xiong et al. [38] also presented an improved version of CLAS technique against collusion attack for VANETs. Here in this paper, we have represented one another version of Cui et al. technique & proved that the technique is safe under CDH assumption.

III. PRELIMINARIES

A. ELLIPTIC CURVE CRYPTOGRAPHY

Here, an Elliptic curve cryptosystem (ECC) is defined, which is popular due to its capability with respect to processing and communication costs and is utilized in the construction of many cryptographic protocols and secure techniques [21].

Assume that the prime integer p determines the finite field $\mathcal{F}_p$. Let $y^2 = x^3 + cx + d(modp)$ define a collection of elliptic curve points E over $\mathcal{F}_p$ where $c, d \in \mathcal{F}_p$ and $(4c^3 + 27d^2)modp \neq 0$. Then $E/\mathcal{F}_p$ represents an elliptic curve E over $\mathcal{F}_p$. Let O represent the point at infinity. Points

over $E/\mathcal{F}_p$ & O are included in the additive group G of the elliptic curve having generator P & order q.

B. ELLIPTIC CURVE DISCRETE LOGARITHMIC PROBLEM

Let T_1 and T_2 are two points over an elliptic curve $E/\mathcal{F}_p$. Given $T_2 = xT_1$ it is hard to find x.

C. CERTIFICATELESS AGGREGATE SIGNATURE SCHEME (CLAS)

A CLAS system is typically made up of eight algorithms, namely: Setup, Generate Pseudonym, Generate Partial Private Key, Generate Vehicle Key, Sign, Verify, Aggregate, and Aggregate Verify.

1) **Setup:** Here TRA and KGC demonstrate this approach by accepting a security parameter κ to create a master public key & master secret key pair (mpk, msk) together with certain system settings.
2) **Generate Pseudonym:** Here TRA employs this technique, which adapts the car's actual identity (RID) and uses it to calculate a pseudo-identity I_D, where the true identity is what makes the car unique.
3) **Generate Partial Private Key:** Here KGC uses this process, accepting a vehicle's pseudo-identity I_{D_i} in order to calculate the vehicle's partial private key, $psk_{I_{D_i}}$, using its master secret key msk_{KGC}. It is believed that the matching car will receive the $psk_{I_{D_i}}$ safely.
4) **Generate Vehicle Key:** The vehicle uses this technique, which generates a vehicle's public and secret key pair $(vpk_{I_{D_i}}, vsk_{I_{D_i}})$ after receiving the vehicle's pseudo-identity (I_{D_i}) as input & choosing a random integer $y_i \in Z_q^*$.
5) **Sign:** Every vehicle V_{H_i} uses this method, which converts a message $\mathfrak{m}_i \in \{0, 1\}^*$ & a signing key $(psk_{I_{D_i}}, vsk_{I_{D_i}})$ into a certificateless signature $\sigma_i = Sign(psk_{I_{D_i}}, vsk_{I_{D_i}}, \mathfrak{m}_i)$.
6) **Verify:** This takes the master public key mpk_{KGC}, the pseudo-identity I_{D_i}, and the corresponding public key $vpk_{I_{D_i}}$ of the vehicle V_{H_i}, & the certificateless signature on message $\mathfrak{m}_i$ as inputs & outputs true or false depending upon the validity of the certificateless signature.
7) **Aggregate:** An aggregate signature generator can combine a number of distinct certificateless signatures by using this approach. When n vehicles $V_{H_1}, V_{H_2}, \ldots, V_{H_n}$ are aggregated into a set $\mathcal{V}$, the pseudo identity I_{D_i} of each the aggregate signature generator calculates $S = \Sigma_{i=1}^{n} S_i$ and outputs $\sigma = (R_{x_1}, R_{x_2}, \ldots, R_{x_n}, S)$ as a certificateless aggregate signature given the vehicle V_{H_i}, the corresponding public key $vpk_{I_{D_i}}$ of V_{H_i}, and message-signature pairs $(\mathfrak{m}_1, \sigma_1 = (R_{x_1}, S_1)), (\mathfrak{m}_2, \sigma_2 = (R_{x_2}, S_2)), \ldots, (\mathfrak{m}_n, \sigma_n = (R_{x_n}, S_n))$ from $\mathcal{V}$, respectively. It is expected that every RSU in our system runs this method.
8) **Aggregate Verify:** The pseudo identity I_{D_i} of each vehicle V_{H_i}, the corresponding public key $vpk_{I_{D_i}}$ of V_{H_i}, and the certificateless aggregate signature σ on messages $(\mathfrak{m}_1, \mathfrak{m}_2, \ldots, \mathfrak{m}_n)$ are the inputs. The output is true if the certificateless aggregate signature is valid, & false otherwise. The aggregating set $\mathcal{V}$ of n vehicles $(V_{H_1}, V_{H_2}, \ldots, V_{H_n})$ is the input.

D. SYSTEM MODEL

VANETs operate well with a network paradigm having two layers, according to earlier studies [15], [35], [41]. We present a model having two layers for vehicle networks. Vehicles on roadways and RSUs at the sides of the roads make up the lower layer. The dedicated short-range communications (DSRC) protocol [19] states that 5.9-GHz DSRC, also known as IEEE 802.11p, is the communication channel utilized between OBUs and RSUs. Every vehicle has a unique actual identity, pseudo-identity, partial private key, public key, and secret key, which are used to sign all communications before forwarding them to the nearby RSU. It is the responsibility of the RSUs receiving traffic-related information to confirm the messages' digital signatures. The Key Generation Center (KGC), the Trace Authority (TRA), & application servers (such as traffic management and analysis centers) make up the upper layer of the vehicular sensor network. Since the KGC & TRA are in charge of system initialization in our approach, we make the assumption that they are reliable and have enough memory, storage, and processing power. The KGC & TRA can be seen as one Trusted Authority (TA) or created jointly, depending on the implementation environment. It is presumed that there is secure communication between TAs and RSUs because these devices are often connected by wired networks that use secure protocols like the Transport Layer Security protocol.

It is assumed that the following is true:

- The KGC and TRA are reliable and rigorous consistently. Reasonably, we presume that two different TAs are separated. Additionally, they have enough processing and storage power. The computing capability of the OBUs is constrained. However, compared to OBUs, RSUs have more processing power.
- Every vehicle has a dependable positioning system, such as a global position system, and it may use GPS to get exact time information. Vehicles adopt this system to compare the message sender's physical location with the location data contained within the identifying string of the RSU.
- Every vehicle has a tamper-proof device installed, which serves as a safe barrier to prevent any data from being removed from the device.

E. UNIVERSAL FORGERY ATTACK

For a universal forgery attack, we consider an adversary who does not have access to the master secret key of the KGC, nor does he have the ability to replace the user's public key or secret key. The adversary has access to signing Oracle over any message of his choice. Adversary successfully models the universal forgery attack if it is able to forge a legitimate

aggregate signature from a single signature in which some signatures are not the output of a signing oracle.

F. SECURITY REQUIREMENTS

A workable method needs to accomplish a number of security attributes, such as the following:

1) **Authentication:** An attestation RSU needs to confirm that a message it receives is from a real user and that harmful editing hasn't been done to it.
2) **Anonymity:** Vehicles communicate with one another using the pseudonym technique to protect privacy and prevent identity information from being disclosed.
3) **Unlinkability:** The attacker should not associate two or more messages with the identical signer, and no malicious vehicle, RSU or adversary should associate two signatures with the identical signer.
4) **Traceability:** In order to understand the facts and prevent mistakes in judgment, the TM uses matching pseudonyms to track the driving records of every vehicle.
5) **Resistance to attacks:** The suggested system needs to be able to fend off well-known attacks like replay and modification.

IV. REVISITING THE CLAS TECHNIQUE OF CUI ET AL.

Cui et al. [9] presented a CLS system & subsequently developed an effective CLAS strategy by building upon the novel CLS scheme. The CLS & CLAS techniques by Cui et al. are briefly reviewed here.

A. CUI ET AL.'S CLS TECHNIQUE

The following five algorithms make up the scheme, which has four parties: Trace Authority (TRA), Key Generation Center (KGC), Signer, & Verifier.

1) **Setup:** The TRA and KGC operate this algorithm as follows:
   - The TRA and KGC select an elliptic curve E described as $y^2 = x^3 + cx + d(modp)$ where $c, d \in \mathcal{F}_p$ and two large prime numbers, p & q, given a security parameter κ. Moreover, P is a point on E that joins to form group G of order q.
   - The KGC chooses an arbitrary $\eta \in Z_q^*$ and evaluates $P_{pub} = \eta P$ while the TRA selects an arbitrary $\gamma \in Z_q^*$ and evaluates $T_{pub} = \gamma P$, where the master keys for traceability and partial key extraction, respectively, are denoted by η and γ.
   - Three safe one-way hash functions are selected by the KGC and TRA $\mathfrak{h}_1 : G \rightarrow Z_q^*$, $\mathfrak{h}_2, \mathfrak{h}_3 : \{0, 1\}^* \rightarrow Z_q^*$. The system parameter $P_{ARM} = (P, p, q, E, G, \mathfrak{h}_1, \mathfrak{h}_2, \mathfrak{h}_3, P_{pub}, T_{pub})$ is released by KGC and TRA.
2) **Generate Pseudo Identity/ Generate Partial Private Key:** Vehicles, TRA, and KGC all executes this algorithm as follows:
   - Once a random $k_\delta \in Z_q^*$ has been chosen, a vehicle V_{H_δ} computes $\mathcal{ID}_{\delta,1} = k_\delta P$ and sends $(RID_\delta, \mathcal{ID}_{\delta,1})$ to the TRA, where RID_δ is $V'_{H_\delta}s$ unique identity.
   - Regarding this pair, the TRA computes $\mathcal{ID}_{\delta,2} = RID_\delta \oplus \mathfrak{h}_\delta(\gamma, \mathcal{ID}_{\delta,1}, \mathcal{T}_\delta, T_{pub})$ after first determining whether RID_δ is valid, where $\mathcal{T}_\delta$ is the pseudo-identity's validity time.
   - The TRA forwards the pseudo-identity of vehicle $V_{H_\delta}, I_{D_\delta} = (\mathcal{ID}_{\delta,1}, \mathcal{ID}_{\delta,2}, \mathcal{T}_\delta)$ to KGC. The partial private key of V_{H_δ} is generated as $PSK_{I_{D_\delta}} = d_\delta + \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}}) \times \eta \ modq$ by the KGC once it receives I_{D_δ}. It does this by choosing a random $d_\delta \in Z_q^*$ and computing $Q_{I_{D_\delta}} = d_\delta P$.
   - Ultimately, the tuple $(Q_{I_{D_\delta}}, PSK_{I_{D_\delta}})$ is securely sent to V_{H_δ}.
3) **Vehicle Key Generation:** Vehicle V_{H_δ} determines its public key as $VPK_{I_{D_\delta}} = \chi_{I_{D_\delta}} \cdot P$ & its secret key as a random number, $\chi_{I_{D_\delta}} = VSK_{I_{D_\delta}} \in Z_q^*$.
4) **Sign:** Given a signing key $(PSK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$ & a traffic related message $\mathfrak{m}_\delta$, vehicle V_{H_δ} does the following:
   - It starts by selecting a random $r_\delta \in Z_q^*$ and computing $R_{x_\delta} = r_\delta P$, $\mathfrak{h}_\delta = \mathfrak{h}_3(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ and $S_\delta = \mathfrak{h}_\delta \cdot r_\delta + PSK_{I_{D_\delta}} modq$.
   - Further, it outputs $\sigma_\delta = (R_{x_\delta}, S_\delta)$ as a certificateless signature on message $\mathfrak{m}_\delta$, where t_δ is the most recent timestamp.
5) **Verify:** The RSU or application server verifies that t_δ is legitimate after getting the message $(I_{D_\delta}, VPK_{I_{D_\delta}}, \mathfrak{m}_\delta, t_\delta, \sigma_\delta)$ from V_{H_δ}. Then, it does the following actions:
   - First, it evaluates $\mathfrak{h}_{\delta,0} = \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}})$ and $\mathfrak{h}_\delta = \mathfrak{h}_3(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$
   - Then, it determines if $S_\delta \cdot P = \mathfrak{h}_\delta \cdot R_{x_\delta} + Q_{I_{D_\delta}} + \mathfrak{h}_{\delta,0} \cdot P_{pub}$ stays true. The message is accepted if it is; if not, it is declined.

B. CUI ET AL.'S CL-AS TECHNIQUE

The algorithms for the Setup, Generate Pseudo Identity/Generate Partial Private Key, Vehicle Key Generation, and Sign are similar to those for the CLS technique mentioned above. Furthermore, the subsequent 2 algorithms are executed.

1) **Aggregate:** Different certificateless signatures can be combined by any RSU, which functions as an aggregator. This process aggregates a batch of n vehicles $\{V_{H_1}, V_{H_2}, \ldots\ldots, V_{H_n}\}$ having pseudo-identities $\{I_{D_1}, I_{D_2}, \ldots\ldots, I_{D_n}\}$ & the message-signature sets $\{(\mathfrak{m}_1, \mathcal{T}_1, \sigma_1 = (R_{x_1}, S_1)), (\mathfrak{m}_2, \mathcal{T}_2, \sigma_2 = (R_{x_2}, S_2)), \ldots\ldots, (\mathfrak{m}_n, \mathcal{T}_n, \sigma_n = (R_{x_n}, S_n))\}$. The RSU computes $S = \Sigma_{\delta=1}^{n} S_\delta$ and returns $\sigma = (R_{x_1}, R_{x_2}, \ldots\ldots, R_{x_n}, S)$ as an aggregate signature.
2) **Aggregate Verify:** The RSU or application server (AS) runs this algorithm. Once the AS or RSU has received the CLAS, $\sigma = (R_{x_1}, R_{x_2}, \ldots\ldots, R_{x_n}, S)$, they verify if the timestamp $\mathcal{T}_\delta$ where $\delta = 1, 2, \ldots\ldots, n$ satisfies the criteria. It performs the following actions if it holds; if not, it aborts.

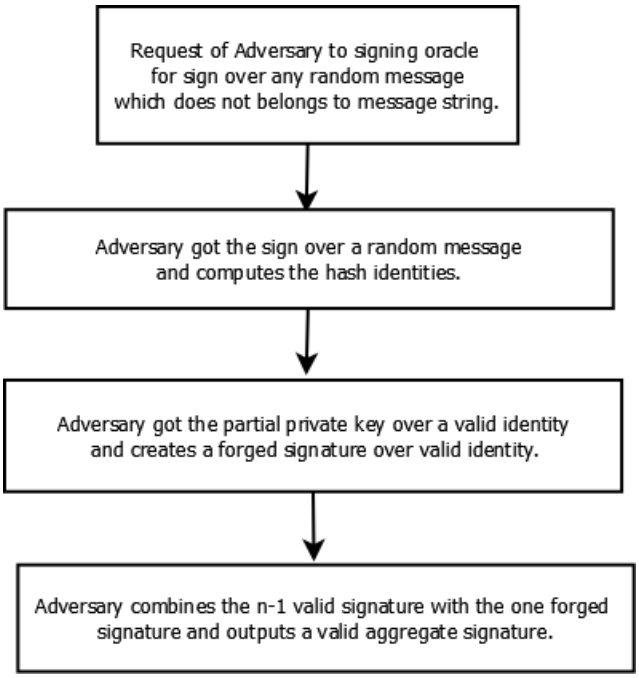

FIGURE 1. Attack Methodology.

- It computes $\mathfrak{h}_{\delta,0} = \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}})$ and $\mathfrak{h}_\delta = \mathfrak{h}_3(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$.
- Check whether $S \cdot P = \Sigma_{\delta=1}^{n}(\mathfrak{h}_\delta \cdot R_{x_\delta}) + \Sigma_{\delta=1}^{n} Q_{I_{D_\delta}} + (\Sigma_{\delta=1}^{n}\mathfrak{h}_{\delta,0}) \cdot P_{pub}$ holds or not. The signature is accepted if it holds; if not, it is denied.

V. FORGERY ATTACK ON CUI ET AL.'S CERTIFICATELESS TECHNIQUE

Cui et al. [9] asserted that, under the ECDLP assumption, their CLS & CL-AS methods are theoretically safe against forgery. Unfortunately, that is untrue. We discovered and provided a specific attack example for the security of Cui et al.'s [9] technique, which is too easily cracked. The next steps are attainable by any honest but curious adversary ($\mathcal{A}$).

- $\mathcal{A}$ picks a singer at random. Without losing generality, we assume $\mathcal{A}$ chooses $\mathfrak{U}_\mathfrak{n}$ with identity I_{D_n} & picks $\mathfrak{m}'_n \in \{0,1\}^*$ ($\mathfrak{m}'_n \neq \mathfrak{m}_\delta, \delta = 1, 2, \ldots, n$). $\mathcal{A}$ requests the signature of I_{D_n} on $\mathfrak{m}'_n$ from the Sign oracle. Consider $\sigma'_n = (R'_{x_n}, S'_n)$ to be the Sign oracle's output.
- Now $\mathcal{A}$ computes
  $\mathfrak{h}'_n = \mathfrak{h}_3(\mathfrak{m}'_n, I_{D_n}, VPK_{I_{D_n}}, R'_{x_n}, t_n)$ and $\mathfrak{h}_2(I_{D_n}, Q_{I_{D_n}})$
  So $\mathcal{A}$ has the equation
  $S'_n = \mathfrak{h}'_n \cdot r'_n + PSK_{I_{D_n}} mod q$ where $PSK_{I_{D_n}} = d_n + \mathfrak{h}_2(I_{D_n}, Q_{I_{D_n}}) \times \eta$
  Therefore $\mathcal{A}$ has
  $S'_n = \mathfrak{h}'_n \cdot r'_n + d_n + \mathfrak{h}_2(I_{D_n}, Q_{I_{D_n}}) \times \eta \; mod q$
  Since $S'_n \cdot P = \mathfrak{h}'_n \cdot r'_n \cdot P + d_n \cdot P + \mathfrak{h}_2(I_{D_n}, Q_{I_{D_n}}) \times \eta \cdot P mod q$
  And $r'_n \cdot P = R'_{x_n}$ and $\eta \cdot P = P_{pub}$
  So $\mathcal{A}$ will easily get $d_n \cdot P = S'_n \cdot P - \mathfrak{h}'_n \cdot R'_{x_n} - \mathfrak{h}_2(I_{D_n}, Q_{I_{D_n}}) \times \eta \cdot P \; mod q$ from which adversary can easily calculate d_n.
  Now adversary randomly chooses $r_n \in Z_q^*$ and computes $R_{x_n} = r_n P, \mathfrak{h}_n = \mathfrak{h}_3(m_n, I_{D_n}, VPK_{I_{D_n}}, R_{x_n}, t_n)$ and $S_n = \mathfrak{h}_n \cdot r_n + d_n + \mathfrak{h}_2(I_{D_n}, Q_{I_{D_n}}) \times \eta mod q$. Here, $\sigma_n = (R_{x_n}, S_n)$ is not the Sign oracle's output; rather, it is a forged signature of I_{D_n} on $\mathfrak{m}_n$.
- $\mathcal{A}$ queries the Sign oracle with the signature of $\mathfrak{m}_\delta$ for $I_{D_\delta}(\delta = 1, 2, \ldots, n-1)$. Let $\sigma_\delta = (R_{x_\delta}, S_\delta)(\delta = 1, 2, \ldots, n-1)$ be the result of the Sign oracle.
- $\mathcal{A}$ calculates $S^* = \Sigma_{\delta=1}^{n} S_\delta$ and generates the aggregate signature $\sigma^* = (R_{x_1}, \ldots, R_{x_n}, S^*)$.
  The verification equation given below consistently holds, indicating that σ^* is a valid aggregate signature.
  $S^* \cdot P = \Sigma_{\delta=1}^{n}(\mathfrak{h}_\delta \cdot R_{x_\delta}) + \Sigma_{\delta=1}^{n} Q_{I_{D_\delta}} + (\Sigma_{\delta=1}^{n}\mathfrak{h}_{\delta,0}) \cdot P_{pub}$.
  Therefore, for an honest but curious KGC, the technique is vulnerable to universal forgery.

Countermeasures for attack resilience:

1) The risk of attack can be reduced with the introduction of some extra hash function and applying it over some key by which it'll be hard for any adversary to calculate it and introduce that in the signing key.
2) The scheme will become more secure if we introduce some extra private key to the user and utilize it in the formation of the signing key.

VI. OUR PROPOSED SCHEME

A. OUR NEW CLS TECHNIQUE

Our new CLS system comprises five algorithms, namely: Setup, Generate Pseudo Identity/Generate Partial Private Key, Vehicle Key Generation, Sign, and Verify. The Setup, Generate Pseudo Identity/Generate Partial Private Key algorithms are almost similar to that of the CLS scheme by Cui et al. mentioned above. The following algorithms are demonstrated below:

1) **Setup:** The TRA & KGC run this algorithm as follows:
   - The TRA and KGC select an elliptic curve E defined by $y^2 = x^3 + cx + d(mod p)$ where $c, d \in \mathcal{F}_p$ and two large prime numbers, p & q, given a security parameter κ. Moreover, P is a point on E that joins to form group G of order q.
   - The KGC chooses an arbitrary $\eta \in Z_q^*$ and evaluates $P_{pub} = \eta P$ while the TRA selects an arbitrary $\gamma \in Z_q^*$ and evaluates $T_{pub} = \gamma P$, where the master keys for traceability and partial key extraction, respectively, are denoted by η and γ.
   - Five safe one-way hash functions are selected by the KGC and TRA $\mathfrak{h}_1, \mathfrak{h}_2, \mathfrak{h}_3 : G \rightarrow Z_q^*$, $\mathfrak{h}_4, \mathfrak{h}_5 : \{0,1\}^* \times G \rightarrow Z_q^*$. The system parameter $P_{ARM} = (P, p, q, E, G, \mathfrak{h}_1, \mathfrak{h}_2, \mathfrak{h}_3, \mathfrak{h}_4, \mathfrak{h}_5, P_{pub}, T_{pub})$ is published by KGC and TRA.
2) **Generate Pseudo Identity/Generate Partial Private Key:** Vehicles, TRA, and KGC all executes this algorithm as follows:
   - Once a random $k_\delta \in Z_q^*$ has been chosen, a vehicle V_{H_δ} computes $\mathcal{ID}_{\delta,1} = k_\delta P$ and sends $(RID_\delta, \mathcal{ID}_{\delta,1})$ to the TRA, where RID_δ is $V'_{H_\delta}s$ unique identity.

- Regarding this pair, the TRA computes $\mathcal{ID}_{\delta,2} = RID_\delta \oplus \mathfrak{h}_1(\gamma, \mathcal{ID}_{\delta,1}, T_\delta, T_{pub})$ after first determining whether RID_δ is valid, where T_δ is the pseudo-identity's validity time.
- The TRA sends to the KGC the pseudo-identity of vehicle V_{H_δ}, $I_{D_\delta} = (\mathcal{ID}_{\delta,1}, \mathcal{ID}_{\delta,2}, T_\delta)$. The partial private key of V_{H_δ} is generated as $PSK_{I_{D_\delta}} = d_\delta + \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}}) \times \eta \ modq$ by the KGC once it receives I_{D_δ}. It does this by choosing a random $d_\delta \in Z_q^*$ and computing $Q_{I_{D_\delta}} = d_\delta P$.
- Ultimately, the tuple $(Q_{I_{D_\delta}}, PSK_{I_{D_\delta}})$ is securely sent to V_{H_δ}.

TABLE 1. Notations and their description.

Notation	Description
V_{H_δ}	δ_{th} vehicle
p,q	Large prime numbers
E	An elliptic curve: $y^2 = x^3 + cx + d(modp)$
G	Additive group with order q
P	Group G's generator
(P_{pub}, η)	KGC's public & private key
(T_{pub}, γ)	TRA's public & private key
PSK_δ	Vehicle $V'_{H_\delta}s$ partial private key
(VPK_δ, VSK_δ)	Vehicle $V'_{H_\delta}s$ public and secret key
RID	Vehicle $V'_{H_\delta}s$ real identity
I_{D_δ}	$V'_{H_\delta}s$ anonymous identity such as $I_{D_\delta} = (\mathcal{ID}_{\delta,1}, \mathcal{ID}_{\delta,2}, T_\delta)$
$\mathfrak{h}_1, \mathfrak{h}_2, \mathfrak{h}_3, \mathfrak{h}_4, \mathfrak{h}_5$	Five secure hash functions
$\mathcal{T}_\delta$	Pseudo identity's valid period
t_δ	The latest time stamp

3) **Vehicle Key Generation:** The vehicle V_{H_δ} selects two confidential values, $\chi^1_{I_{D_\delta}}$ and $\chi^2_{I_{D_\delta}}$ and set its secret key as $VSK_{I_{D_\delta}} = (\chi^1_{I_{D_\delta}} + \chi^2_{I_{D_\delta}})$ and evaluates its public key as $VPK_{I_{D_\delta}} = (\chi^1_{I_{D_\delta}} + \chi^2_{I_{D_\delta}}) \cdot P$. Also it takes $VPK^1_{I_{D_\delta}} = \chi^1_{I_{D_\delta}} \cdot P$ and $VPK^2_{I_{D_\delta}} = \chi^2_{I_{D_\delta}} \cdot P$
4) **Sign:** Given a signing key $(PSK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$ and a traffic related message $\mathfrak{m}_\delta$, vehicle V_{H_δ} does the following:
   - It starts by selecting a random $r_\delta \in Z_q^*$ & evaluates $R_{x_\delta} = r_\delta P$, $\mathfrak{h}_\delta = \mathfrak{h}_3(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ and $S_\delta = \mathfrak{h}_\delta \cdot r_\delta + PSK_{I_{D_\delta}} modq + \chi^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1} + \chi^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2}$ where $\mathfrak{h}_{\delta,1} = \mathfrak{h}_4(\mathfrak{m}_\delta, I_{D_\delta}, VPK^1_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ and $\mathfrak{h}_{\delta,2} = \mathfrak{h}_5(\mathfrak{m}_\delta, I_{D_\delta}, VPK^2_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$
   - Further, it outputs $\sigma_\delta = (R_{x_\delta}, S_\delta)$ as a certificateless signature on message $\mathfrak{m}_\delta$, where t_δ is the most recent timestamp.
5) **Verify:** The application server or RSU verifies that t_δ is legitimate after receiving the message $(I_{D_\delta}, VPK_{I_{D_\delta}}, \mathfrak{m}_\delta, t_\delta, \sigma_\delta)$ from V_{H_δ}. Then, it does the following actions:
   - First, it evaluates $\mathfrak{h}_{\delta,0} = \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}})$ and $\mathfrak{h}_\delta = \mathfrak{h}_3(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$, $\mathfrak{h}_{\delta,1} = \mathfrak{h}_4(\mathfrak{m}_\delta, I_{D_\delta}, VPK^1_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ and $\mathfrak{h}_{\delta,2} = \mathfrak{h}_5(\mathfrak{m}_\delta, I_{D_\delta}, VPK^2_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$.
   - Then, it determines if $S_\delta \cdot P = \mathfrak{h}_\delta \cdot R_{x_\delta} + Q_{I_{D_\delta}} + \mathfrak{h}_{\delta,0} \cdot P_{pub} + VPK^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1} + VPK^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2}$ stays true. The message is accepted if it is; if not, it is rejected.

Correctness: Since $P_{pub} = \eta \cdot P$, $Q_{R_{x_\delta}} = d_\delta P$, $PSK_{I_{D_\delta}} = d_\delta + \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}}) \times \eta modq$, $R_{x_\delta} = r_\delta P$ *and* $S_\delta = \mathfrak{h}_\delta \cdot r_\delta + PSK_{I_{D_\delta}} modq + \chi^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1} + \chi^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2}$.

We obtain $S_\delta \cdot P = (\mathfrak{h}_\delta \cdot r_\delta + PSK_{I_{D_\delta}} modq + \chi^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1} + \chi^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2}) \cdot P$

$= \mathfrak{h}_\delta \cdot r_\delta \cdot P + (d_\delta + \mathfrak{h}_{\delta,0} \times \eta) \cdot P + \chi^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1} \cdot P + \chi^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2} \cdot P$

$= \mathfrak{h}_\delta \cdot R_{x_\delta} + Q_{I_{D_\delta}} + \mathfrak{h}_{\delta,0} \cdot P_{pub} + VPK^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1} + VPK^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2}$.

B. OUR NEW CLAS SCHEME

The algorithms for the Setup, Generate Pseudo Identity/Generate Partial Private Key, Vehicle Key Generation, and Sign are similar to those for the CLS technique mentioned above. Furthermore, the subsequent two algorithms are executed.

1) **Aggregate:** Different certificateless signatures can be combined by any RSU, which functions as an aggregator. This process aggregates a set of n vehicles $\{V_{H_1}, V_{H_2}, \ldots\ldots, V_{H_n}\}$ with pseudo-identities $\{I_{D_1}, I_{D_2}, \ldots\ldots, I_{D_n}\}$ & the message-signature pairs $\{(\mathfrak{m}_1, \mathcal{T}_1, \sigma_1 = (R_{x_1}, S_1)), (\mathfrak{m}_2, \mathcal{T}_2, \sigma_2 = (R_{x_2}, S_2)), \ldots\ldots, (\mathfrak{m}_n, \mathcal{T}_n, \sigma_n = (R_{x_n}, S_n))\}$. The RSU evaluates $S = \Sigma_{\delta=1}^n S_\delta$ & returns $\sigma = (R_{x_1}, R_{x_2}, \ldots\ldots, R_{x_n}, S)$ as an aggregate signature.
2) **Aggregate Verify:** The RSU or application server (AS) runs this algorithm. Once the AS or RSU has received the CLAS, $\sigma = (R_{x_1}, R_{x_2}, \ldots\ldots, R_{x_n}, S)$, they verify if the timestamp $\mathcal{T}_\delta$ where $\delta = 1, 2, \ldots\ldots, n$ satisfies the criteria. It performs the following actions if it holds; if not, it aborts.
   - It computes $\mathfrak{h}_{\delta,0} = \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}})$ and $\mathfrak{h}_\delta = \mathfrak{h}_3(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$, $\mathfrak{h}_{\delta,1} = \mathfrak{h}_4(\mathfrak{m}_\delta, I_{D_\delta}, VPK^1_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ and $\mathfrak{h}_{\delta,2} = \mathfrak{h}_5(\mathfrak{m}_\delta, I_{D_\delta}, VPK^2_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$.
   - Check whether $S \cdot P = \Sigma_{\delta=1}^n(\mathfrak{h}_\delta \cdot R_{x_\delta}) + \Sigma_{\delta=1}^n Q_{I_{D_\delta}} + (\Sigma_{\delta=1}^n \mathfrak{h}_{\delta,0}) \cdot P_{pub} + \Sigma_{\delta=1}^n(VPK^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1}) + \Sigma_{\delta=1}^n(VPK^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1})$ holds or not. The signature is accepted if it holds; if not, it is denied.

Correctness: Since $P_{pub} = \eta \cdot P$, $Q_{I_{D_\delta}} = d_\delta P$, $PSK_{I_{D_\delta}} = d_\delta + \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}}) \times \eta modq$, $R_{x_\delta} = r_\delta P$ *and* $S_\delta = \mathfrak{h}_\delta \cdot r_\delta + PSK_{I_{D_\delta}} modq + \chi^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1} + \chi^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2}$ and $S = \Sigma_{\delta=1}^n S_\delta$.

We obtain $S \cdot P = \Sigma_{\delta=1}^n S_\delta \cdot P$

$= \Sigma_{\delta=1}^n(\mathfrak{h}_\delta \cdot r_\delta + PSK_{I_{D_\delta}} modq + \chi^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1} + \chi^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2}) \cdot P$

$= \Sigma_{\delta=1}^n(\mathfrak{h}_\delta \cdot r_\delta) \cdot P + \Sigma_{\delta=1}^n(d_\delta + \mathfrak{h}_{\delta,0} \times \eta) \cdot P + \Sigma_{\delta=1}^n(\chi^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1}) \cdot P + \Sigma_{\delta=1}^n(\chi^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2}) \cdot P$

$= \Sigma_{\delta=1}^n(\mathfrak{h}_\delta \cdot R_{x_\delta}) + \Sigma_{\delta=1}^n(d_\delta \cdot P) + \Sigma_{\delta=1}^n(\mathfrak{h}_{\delta,0}) \cdot \eta \cdot P + \Sigma_{\delta=1}^n(VPK^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1}) + \Sigma_{\delta=1}^n(VPK^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2})$

$= \Sigma_{\delta=1}^n(\mathfrak{h}_\delta \cdot R_{x_\delta}) + \Sigma_{\delta=1}^n Q_{I_{D_\delta}} + (\Sigma_{\delta=1}^n \mathfrak{h}_{\delta,0}) \cdot P_{pub} + \Sigma_{\delta=1}^n(VPK^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1}) + \Sigma_{\delta=1}^n(VPK^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1})$

- The scheme by Cui et al. was not secure against a universal forgery attack since an adversary was able to forge a legitimate aggregate signature from a set of signatures which are not the output of the signing oracle; instead, some of them are forged ones. With the introduction of one extra hash function and secret key, it becomes hard for an adversary to get a forged signature over any message of his choice. In the signing algorithm of our proposed scheme, the signature we have proposed is extra secure, and any adversary fails to output a forged signature over any message of his choice without any secret information of the user. Thus, our proposed scheme is secure against the attack we have mounted on Cui et al.'s scheme.
- Previously, Kamil et al. also reviewed the scheme by Cui et al., and they mounted a Type II adversary attack over the CLS and CLAS scheme by Cui et al. Here, we have shown that Cui et al.'s scheme does not hold universal forgery attacks, and we have reviewed the scheme by Cui et al. In our next section, we'll be giving the security proof of our new scheme against universal forgery attacks and Type II adversaries. Thus, our new scheme is secure against universal forgery attacks as well as Type II adversary attacks that were mounted by Kamil et al. over Cui et al.'s scheme.

VII. SECURITY PROOF

This section evaluates the security of our presented CLS & CLAS schemes for VANETs. We demonstrate that the suggested technique can ensure non-forgery under the random oracle model (ROM), presuming that the computational ECDLP is hard.

- **Definition 1. Forking Lemma:** Consider A, a probabilistic polynomial-time Turing machine that only accepts public data as input. Q & R represent the no. of queries that A can ask the random oracle & the signer, respectively. Assume A generates a valid signature $(\mathfrak{m}, \sigma_1, h, \sigma_2)$ with probability $\varepsilon \geq 10(R+1)(R+Q)/2^k$ within a given time T. Without the knowledge of the secret key, if the triples (σ_1, h, σ_2) get simulated with a comparable distribution probability, then the machine obtained from A is controlled by another machine substituting interaction with the signer by simulating & creates 2 valid signatures $(\mathfrak{m}, \sigma_1, h, \sigma_2)$ & (σ_1, h, σ_2) such that $h \neq h'$ in expected time $T' \geq 120686QT/\varepsilon$.

A. CERTIFICATELESS SIGNATURE (CLS)

Theorem 1: Based on the ROM, the suggested CLS approach is existentially unforgeable against an adaptive chosen message.

The theorem derives directly from Lemma 1 according to Forking Lemma.

Lemma 1: A probabilistic polynomial-time forger, $\mathcal{A}_1$, may have an advantage in forging a CLS in an attack modelled by Game 1 of Definition 1 after performing $Q_{\mathfrak{h}_\delta}$ questions to random oracles $\mathfrak{h}_\delta$ for $\delta = 2, 3, 4, 5$, Q_{Ppk}, Q_{CV}, Q_{SK} and Q_{Sign} inquiries to the Partial-Private-Key oracle, Create-Vehicle oracle, Secret-Key oracle and Sign oracle respectively, then there is a challenger C_1 who can solve ECDLP within a time period T which is projected to be less than 120686QT/ε if $\varepsilon \geq 10(Q_{Sig}+1)\,(Q_{\mathfrak{h}_2}+Q_{\mathfrak{h}_3}+Q_{\mathfrak{h}_4}+Q_{\mathfrak{h}_5}+Q_{Ppk}+Q_{CV}+Q_{SK}+Q_{Sign})/q$.

Proof. Let us given an ECDLP sample $(T_1, T_2 = xT_1)$ where T_1 and T_2 are two points on elliptic curve E & an adversary $\mathcal{A}_1$ could forge message $(I_{D_\delta}, VPK_\delta, \mathfrak{m}_\delta, t_\delta, \sigma_\delta)$. Set up a game between $\mathcal{A}_1$ & a challenger C_1. C_1 can solve the ECDLP by running $\mathcal{A}_1$ as a subroutine with a probability greater than zero.

- Setup: An arbitrary number η is taken by C_1 and set it as its master key. It further evaluates the corresponding public key as $P_{pub} = \eta P$. To $\mathcal{A}_1$, C_1 forwards the system parameter $P_{ARM} = (P, p, q, E, G, \mathfrak{h}_1, \mathfrak{h}_2, \mathfrak{h}_3, \mathfrak{h}_4, \mathfrak{h}_5, P_{pub}, T_{pub})$.
- $\mathfrak{h}_2$ hash query: When $\mathfrak{h}_2$ query is made by $\mathcal{A}_1$ on the tuple $(I_{D_\delta}, Q_{I_{D_\delta}})$, C_1 checks the list $L_{\mathfrak{h}_2}$ and examines the availability of the tuple in the hash record. If so, it returns $\tau_{\mathfrak{h}_2} = \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}})$ to $\mathcal{A}_1$. Else C_1 chooses an arbitrary $\tau_{\mathfrak{h}_2} \in Z_q^*$ adds it into the hash record $L_{\mathfrak{h}_2}$ & forwards it to $\mathcal{A}_1$.
- $\mathfrak{h}_3$ hash query: When $\mathfrak{h}_3$ query is made by $\mathcal{A}_1$ on the tuple $(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$, C_1 checks the list $L_{\mathfrak{h}_3}$ and examines the availability of the tuple in the hash list. If so, it returns $\tau_{\mathfrak{h}_3} = \mathfrak{h}_3(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ to $\mathcal{A}_1$. Else C_1 selects an arbitrary $\tau_{\mathfrak{h}_3} \in Z_q^*$ includes it into the hash list $L_{\mathfrak{h}_3}$ and forwards it to $\mathcal{A}_1$.
- $\mathfrak{h}_4$ hash query: When $\mathfrak{h}_4$ query is made by $\mathcal{A}_1$ on the tuple $(\mathfrak{m}_\delta, I_{D_\delta}, VPK^1_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$, C_1 checks the list $L_{\mathfrak{h}_4}$ and examines the availability of the tuple in the hash list. If so, it returns $\tau_{\mathfrak{h}_4} = \mathfrak{h}_4(\mathfrak{m}_\delta, I_{D_\delta}, VPK^1_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ to $\mathcal{A}_1$. Else C_1 selects an arbitrary $\tau_{\mathfrak{h}_4} \in Z_q^*$ includes it into the hash list $L_{\mathfrak{h}_4}$ and forwards it to $\mathcal{A}_1$.
- $\mathfrak{h}_5$ hash query: When $\mathfrak{h}_5$ query is made by $\mathcal{A}_1$ on the tuple $(\mathfrak{m}_\delta, I_{D_\delta}, VPK^2_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$, C_1 checks the list $L_{\mathfrak{h}_5}$ and examines the availability of the tuple in the hash list. If so, it returns $\tau_{\mathfrak{h}_5} = \mathfrak{h}_5(\mathfrak{m}_\delta, I_{D_\delta}, VPK^2_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ to $\mathcal{A}_1$. Else C_1 selects an arbitrary $\tau_{\mathfrak{h}_5} \in Z_q^*$ adds it into the hash list $L_{\mathfrak{h}_5}$ and forwards it to $\mathcal{A}_1$.
- Partial private key query: When the partial private key query is made by $\mathcal{A}_1$ on a pseudo-identity I_{D_δ}, C_1 calculates $Q_{I_{D_\delta}} = d_\delta$ and examines if $(I_{D_\delta}, Q_{I_{D_\delta}}, \tau_{\mathfrak{h}_2})$ is present in $L_{\mathfrak{h}_2}$ or not. If C_1 doesn't get the matching tuple $(I_{D_\delta}, Q_{I_{D_\delta}}, \tau_{\mathfrak{h}_2})$, it fails and stops since it is unable to reply the inquiry. Else it evaluates $PSK_{I_{D_\delta}} = d_\delta + \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}}) \times \eta \; mod q$ and sends it to $\mathcal{A}_1$.
- Create Vehicle query: Let us assume that the request relates to the pseudo-identity I_{D_δ}.

  If $(I_{D_\delta}, VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$ is in the list L, C_1 checks if $VPK_{I_{D_\delta}} = \perp$. If $VPK_{I_{D_\delta}} \neq \perp$, C_1 sends $VPK_{I_{D_\delta}}$ to $\mathcal{A}_1$. Otherwise it selects two arbitrary numbers v_{δ_1} and $v_{\delta_2} \in Z_q^*$ and calculates $VPK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})P$ and sets

$VSK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})$. Then C_1 sends $VPK_{I_{D_\delta}}$ to $\mathcal{A}_1$ and updates the list L by $(VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$.

If $(I_{D_\delta}, VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$ is not in the list L, C_1 sets $VPK_{I_{D_\delta}} = \perp$ and it selects two arbitrary numbers v_{δ_1} and $v_{\delta_2} \in Z_q^*$ and evaluates $VPK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})P$ and sets $VSK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})$. Then C_1 sends $VPK_{I_{D_\delta}}$ to $\mathcal{A}_1$ and updates the list L by $(VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$.

- Secret key query: Let us assume that the request relates to the pseudo-identity I_{D_δ}.

  If $(I_{D_\delta}, VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$ is in the list L, C_1 checks if $VSK_{I_{D_\delta}} = \perp$. If $VSK_{I_{D_\delta}} \neq \perp$, C_1 sends $VSK_{I_{D_\delta}}$ to $\mathcal{A}_1$. Otherwise a Create vehicle query is made by C_1 to itself to get $VPK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})P$ and $VSK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})$. Then C_1 forwards $VSK_{I_{D_\delta}}$ to $\mathcal{A}_1$ and updates the list L by $(VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$.

  If $(I_{D_\delta}, VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$ is not present in the record L, a Create vehicle query is requested by C_1 to itself & forwards $VSK_{I_{D_\delta}}$ to $\mathcal{A}_1$ & updates the record L by $(VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$.
- Sign query: When sign query is made by $\mathcal{A}_1$ on message $\mathfrak{m}_\delta$ with pseudo identity I_{D_δ}, C_1 first find the tuple value $(I_{D_\delta}, Q_{I_{D_\delta}}, \tau_{\mathfrak{h}_2})$ from the hash record $L_{\mathfrak{h}_2}$. Then recovers $\tau_{\mathfrak{h}_2}$ from the tuple $(I_{D_\delta}, Q_{I_{D_\delta}}, \tau_{\mathfrak{h}_2})$ and selects two arbitrary numbers r_δ and $\mathfrak{h}_\delta$. Next, C_1 chooses another two arbitrary numbers s_δ and u_δ and attempts again. Additionally, C_1 evaluates $R_{x_\delta} = \mathfrak{h}_\delta^{-1} s_\delta P - Q$ and $S_\delta = s_\delta$, and then forwards (R_{x_δ}, S_δ) to $\mathcal{A}_1$ and adds $(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta, \tau_{\mathfrak{h}_3})$ to the record $L_{\mathfrak{h}_3}$.

  By using Forking lemma [30], after replaying $\mathcal{A}_1$ with identical arbitrary element, C_1 gets two signatures $\sigma_\delta = (R_{x_\delta}, S_\delta)$ and $\sigma'_\delta = (R'_{x_\delta}, S'_\delta)$ which are valid within a polynomial time where $S_\delta = \mathfrak{h}_\delta \cdot r_\delta + Q$ and $S'_\delta = \mathfrak{h}'_\delta \cdot r_\delta + Q$ by computing
  $$\frac{\mathfrak{h}'_\delta S_\delta - \mathfrak{h}_\delta S'_\delta}{\mathfrak{h}'_\delta - \mathfrak{h}_\delta} \; mod q$$
  $$= PSK_{I_{D_\delta}}$$
  As a result C_1 can break the ECDLP within an expected time less than 120686QT/ε if $\varepsilon \geq 10(Q_{Sig} + 1)(Q_{\mathfrak{h}_2} + Q_{\mathfrak{h}_3} + Q_{\mathfrak{h}_4} + Q_{\mathfrak{h}_5} + Q_{Ppk} + Q_{CV} + Q_{SK} + Q_{Sign})/q$. Despite the hardness of solving the ECDLP, the suggested CLS technique can withstand forgery attacks.

Theorem 2: Based on the ROM, the suggested CLS approach is existentially unforgeable against a Type II adversary $\mathcal{A}_2$.

Proof. Let us given an ECDLP sample $(T_1, T_2 = xT_1)$ where T_1 and T_2 are two points on elliptic curve E & an adversary $\mathcal{A}_2$ could forge message $(I_{D_\delta}, VPK_\delta, \mathfrak{m}_\delta, t_\delta, \sigma_\delta)$. Set up a game between $\mathcal{A}_2$ & a challenger C_2. C_1 can solve the ECDLP by running $\mathcal{A}_2$ as a subroutine with a probability greater than zero.

- Setup: An arbitrary number η is taken by C_2 and set it as its master key. It further evaluates the corresponding public key as $P_{pub} = \eta P$. To $\mathcal{A}_2$, C_2 forwards the system parameter $P_{ARM} = (P, p, q, E, G, \mathfrak{h}_1, \mathfrak{h}_2, \mathfrak{h}_3, \mathfrak{h}_4, \mathfrak{h}_5, P_{pub}, T_{pub})$.
- $\mathfrak{h}_2$ hash query: When $\mathfrak{h}_2$ query is made by $\mathcal{A}_2$ on the tuple $(I_{D_\delta}, Q_{I_{D_\delta}})$, C_2 checks the list $L_{\mathfrak{h}_2}$ and examines the availability of the tuple in the hash record. If so, it returns $\tau_{\mathfrak{h}_2} = \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}})$ to $\mathcal{A}_2$. Else C_2 chooses an arbitrary $\tau_{\mathfrak{h}_2} \in Z_q^*$ adds it into the hash record $L_{\mathfrak{h}_2}$ & forwards it to $\mathcal{A}_2$.
- $\mathfrak{h}_3$ hash query: When $\mathfrak{h}_3$ query is made by $\mathcal{A}_2$ on the tuple $(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$, C_2 checks the list $L_{\mathfrak{h}_3}$ and examines the availability of the tuple in the hash list. If so, it returns $\tau_{\mathfrak{h}_3} = \mathfrak{h}_3(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ to $\mathcal{A}_2$. Else C_2 selects an arbitrary $\tau_{\mathfrak{h}_3} \in Z_q^*$ includes it into the hash list $L_{\mathfrak{h}_3}$ and forwards it to $\mathcal{A}_2$.
- $\mathfrak{h}_4$ hash query: When $\mathfrak{h}_4$ query is made by $\mathcal{A}_2$ on the tuple $(\mathfrak{m}_\delta, I_{D_\delta}, VPK^1_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$, C_1 checks the list $L_{\mathfrak{h}_4}$ and examines the availability of the tuple in the hash list. If so, it returns $\tau_{\mathfrak{h}_4} = \mathfrak{h}_4(\mathfrak{m}_\delta, I_{D_\delta}, VPK^1_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ to $\mathcal{A}_2$. Else C_2 selects an arbitrary $\tau_{\mathfrak{h}_4} \in Z_q^*$ includes it into the hash list $L_{\mathfrak{h}_4}$ and forwards it to $\mathcal{A}_2$.
- $\mathfrak{h}_5$ hash query: When $\mathfrak{h}_5$ query is made by $\mathcal{A}_2$ on the tuple $(\mathfrak{m}_\delta, I_{D_\delta}, VPK^2_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$, C_2 checks the list $L_{\mathfrak{h}_5}$ and examines the availability of the tuple in the hash list. If so, it returns $\tau_{\mathfrak{h}_5} = \mathfrak{h}_5(\mathfrak{m}_\delta, I_{D_\delta}, VPK^2_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ to $\mathcal{A}_2$. Else C_2 selects an arbitrary $\tau_{\mathfrak{h}_5} \in Z_q^*$ adds it into the hash list $L_{\mathfrak{h}_5}$ and forwards it to $\mathcal{A}_2$.
- Partial private key query: When the partial private key query is made by $\mathcal{A}_2$ on a pseudo-identity I_{D_δ}, C_2 calculates $Q_{I_{D_\delta}} = d_\delta$ and examines if $(I_{D_\delta}, Q_{I_{D_\delta}}, \tau_{\mathfrak{h}_2})$ is present in $L_{\mathfrak{h}_2}$ or not. If C_2 doesn't get the matching tuple $(I_{D_\delta}, Q_{I_{D_\delta}}, \tau_{\mathfrak{h}_2})$, it fails and stops since it is unable to reply the inquiry. Else it evaluates $PSK_{I_{D_\delta}} = d_\delta + \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}}) \times \eta \; mod q$ and sends it to $\mathcal{A}_2$.
- Create Vehicle query: Let us assume that the request relates to the pseudo-identity I_{D_δ}.

  If $(I_{D_\delta}, VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$ is in the list L, C_2 checks if $VPK_{I_{D_\delta}} = \perp$. If $VPK_{I_{D_\delta}} \neq \perp$, C_2 sends $VPK_{I_{D_\delta}}$ to $\mathcal{A}_2$. Otherwise it selects two arbitrary numbers v_{δ_1} and $v_{\delta_2} \in Z_q^*$ and calculates $VPK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})P$ and sets $VSK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})$. Then C_2 sends $VPK_{I_{D_\delta}}$ to $\mathcal{A}_2$ and updates the list L by $(VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$.

  If $(I_{D_\delta}, VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$ is not in the list L, C_2 sets $VPK_{I_{D_\delta}} = \perp$ and it selects two arbitrary numbers v_{δ_1} and $v_{\delta_2} \in Z_q^*$ and evaluates $VPK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})P$ and sets $VSK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})$. Then C_2 sends $VPK_{I_{D_\delta}}$ to $\mathcal{A}_2$ and updates the list L by $(VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$.
- Secret key query: Let us assume that the request relates to the pseudo-identity I_{D_δ}.

  If $(I_{D_\delta}, VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$ is in the list L, C_2 checks if $VSK_{I_{D_\delta}} = \perp$. If $VSK_{I_{D_\delta}} \neq \perp$, C_2 sends $VSK_{I_{D_\delta}}$ to $\mathcal{A}_2$. Otherwise a Create vehicle query is made by C_2 to itself to get $VPK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})P$ and $VSK_{I_{D_\delta}} = (v_{\delta_1} + v_{\delta_2})$. Then C_2 forwards $VSK_{I_{D_\delta}}$ to $\mathcal{A}_1$ and updates the list L by $(VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$.

  If $(I_{D_\delta}, VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$ is not present in the record L, a Create vehicle query is requested by C_2 to itself

& forwards $VSK_{I_{D_\delta}}$ to $\mathcal{A}_2$ & updates the record L by $(VPK_{I_{D_\delta}}, VSK_{I_{D_\delta}})$.

- Sign query: When sign query is made by $\mathcal{A}_2$ on message $\mathfrak{m}_\delta$ with pseudo identity I_{D_δ}, C_2 first find the tuple value $(I_{D_\delta}, Q_{I_{D_\delta}}, \tau_{\mathfrak{h}_2})$ from the hash record $L_{\mathfrak{h}_2}$. Then recovers $\tau_{\mathfrak{h}_2}$ from the tuple $(I_{D_\delta}, Q_{I_{D_\delta}}, \tau_{\mathfrak{h}_2})$ and selects two arbitrary numbers r_δ and $\mathfrak{h}_\delta$. Next, C_2 chooses another two arbitrary numbers s_δ and u_δ and attempts again. Additionally, C_2 evaluates $R_{x_\delta} = \mathfrak{h}_\delta^{-1} s_\delta P - Q$ and $S_\delta = s_\delta$, and then forwards (R_{x_δ}, S_δ) to $\mathcal{A}_2$ and adds $(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta, \tau_{\mathfrak{h}_3})$ to the record $L_{\mathfrak{h}_3}$.
  By using Forking lemma [30], after replaying $\mathcal{A}_2$ with identical arbitrary element, C_2 gets two signatures $\sigma_\delta = (R_{x_\delta}, S_\delta)$ and $\sigma'_\delta = (R'_{x_\delta}, S'_\delta)$ which are valid within a polynomial time where $S_\delta = \mathfrak{h}_\delta \cdot r_\delta + Q$ and $S'_\delta = \mathfrak{h}'_\delta \cdot r_\delta + Q$ by computing
  $$\frac{\mathfrak{h}'_\delta S_\delta - \mathfrak{h}_\delta S'_\delta}{\mathfrak{h}'_\delta - \mathfrak{h}_\delta} \; modq$$
  $$= PSK_{I_{D_\delta}}$$
  As a result C_2 can break the ECDLP within an expected time less than 120686QT/ε if $\varepsilon \geq 10(Q_{Sig} + 1)\,(Q_{\mathfrak{h}_2} + Q_{\mathfrak{h}_3} + Q_{\mathfrak{h}_4} + Q_{\mathfrak{h}_5} + Q_{Ppk} + Q_{CV} + Q_{SK} + Q_{Sign})/q$. Despite the hardness of solving the ECDLP, the suggested CLS technique is existentially unforgeable against a Type II adversary $\mathcal{A}_2$.

VIII. DISCUSSION

A. SECURITY ANALYSIS

- Message integrity & authentication: Since Theorem 1 states that the ECDLP is hard, no polynomial adversary can create a valid message by forging one. As a result, the verifier can assess the integrity & validity of the message $(I_{D_\delta}, VPK_\delta, \mathfrak{m}_\delta, t_\delta, \sigma_\delta)$ by verifying whether the equation $S_\delta \cdot P = \mathfrak{h}_\delta \cdot R_{x_\delta} + Q_{I_{D_\delta}} + \mathfrak{h}_{\delta,0} \cdot P_{pub} + VPK^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1} + VPK^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2}$ holds, where $\mathfrak{h}_{\delta,0} = \mathfrak{h}_2(I_{D_\delta}, Q_{I_{D_\delta}})$, $\mathfrak{h}_\delta = \mathfrak{h}_3(\mathfrak{m}_\delta, I_{D_\delta}, VPK_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$, $\mathfrak{h}_{\delta,1} = \mathfrak{h}_4(\mathfrak{m}_\delta, I_{D_\delta}, VPK^1_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$ and $\mathfrak{h}_{\delta,2} = \mathfrak{h}_5(\mathfrak{m}_\delta, I_{D_\delta}, VPK^2_{I_{D_\delta}}, R_{x_\delta}, t_\delta)$. Message authentication and integrity are thus provided by the suggested VANET scheme.
- Identity privacy-preserving: The vehicle's pseudo-identity can be calculated by the knowledge of $\gamma \cdot \mathcal{ID}_{\delta,1}$, since pseudo-identity in the proposed technique is the combination of γ and $\mathcal{ID}_{\delta,1}$, where $\mathcal{ID}_{\delta,1}$ is calculated by the vehicle's chosen secret key k_δ. It is impossible for any adversary to deduce the vehicle's true identity from its pseudo identity without knowledge of either γ or k_δ. In other words, the intractability of the CDH problem determines how secure the identification of the vehicle is. Specifically, assuming $T_{pub} = \gamma P$ and $\mathcal{ID}_{\delta,1} = k_\delta P$, compute $\gamma \cdot \mathcal{ID}_{\delta,1} = \gamma k_\delta P$ As a result, using a pseudo-identity, an adversary fails to get any information. Users' ability to preserve their identity and privacy can thus be well maintained.
- Traceability: To get the vehicle's true identity, a request to TRA should be made. TRA can identify a vehicle's genuine identification by using its pseudo-identity $I_{D_\delta} = (\mathcal{ID}_{\delta,1}, \mathcal{ID}_{\delta,2}, T_\delta)$ by the calculation of equation $\mathcal{ID}_{\delta,2} = RID_\delta \oplus \mathfrak{h}_1(\gamma, \mathcal{ID}_{\delta,1}, \mathcal{T}_\delta, T_{pub})$. Therefore, when a vehicle is designated as contentious, the TRA can determine its genuine identity, ensuring traceability.
- Unlinkability: The vehivle V_{H_δ} sends messages $(I_{D_\delta}, VPK_\delta, \mathfrak{m}_\delta, t_\delta, \sigma_\delta)$ to the other people, where $\mathcal{ID}_{\delta,1} = k_\delta P$ and for each message, the random nonce k_δ creates a different anonymous identity for the vehicle. Every signature has a unique, anonymous identity. This specific vehicle isn't able to be associated with any two of the signatures; hence, unlinkability is supported by the suggested approach.
- Resistance to attacks: We provide evidence that the suggested VANET scheme is resistant to impersonation, replay, modification, and collusion attacks. We describe them as follows:

  Impersonation attack: The attacker can impersonate a message if he has the ability to forge a valid message $(I_{D_\delta}, VPK_\delta, \mathfrak{m}_\delta, t_\delta, \sigma_\delta)$ satisfying the verification equation $S_\delta \cdot P = \mathfrak{h}_\delta \cdot R_{x_\delta} + Q_{I_{D_\delta}} + \mathfrak{h}_{\delta,0} \cdot P_{pub} + VPK^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1} + VPK^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2}$. Nevertheless, Theorem 1 indicates that the likelihood of the counterfeit message meeting the verification equation may be low. As a result, the suggested scheme is resistant to this attack.

  Replay attack: The timestamp t_δ is embodied in the message $(I_{D_\delta}, VPK_\delta, \mathfrak{m}_\delta, t_\delta, \sigma_\delta)$.In order to identify the message replay, the RSU & other vehicles will verify if the timestamp t_δ is the current timestamp or not. As a result, the suggested VANET method may withstand replay attacks.

  Modification attack: A digital signature $(I_{D_\delta}, \sigma_\delta)$ is contained in the valid message $(I_{D_\delta}, VPK_\delta, \mathfrak{m}_\delta, t_\delta, \sigma_\delta)$ in our suggested scheme. Thus, for a verifier it is easy to check if someone forges any modification in the valid message $(I_{D_\delta}, VPK_\delta, \mathfrak{m}_\delta, t_\delta, \sigma_\delta)$ by the verification equation $S_\delta \cdot P = \mathfrak{h}_\delta \cdot R_{x_\delta} + Q_{I_{D_\delta}} + \mathfrak{h}_{\delta,0} \cdot P_{pub} + VPK^1_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,1} + VPK^2_{I_{D_\delta}} \cdot \mathfrak{h}_{\delta,2}$. Thus, our scheme is secure against modification attacks.

B. PERFORMANCE EVALUATION

This section examines the computation cost of the suggested method and compares it to some of the previously suggested CL-AS techniques for VANETs.

1) COMPUTATION COSTS

In this study, we use the computation evaluation approach for VANETs presented in [14], which creates a bilinear pairing on the security level of 80 bits as follows: $e : G_1 \times G_2 \rightarrow G_T$, where G_1 symbolises an additive group created by a point P whose order is q on a supersingular elliptic curve $E : y^2 = x^3 + x \; modp$ with degree 2 embedding, where p is a 512-bit

TABLE 2. Different cryptographic operations (Crypt Opr) with their execution time(ET).

Crypt Opr	Time (ms)
T_{BP}	4.2110
T_{m-BP}	1.7090
T_{a-BP}	0.0071
T_{m-ECC}	0.4420
T_{a-ECC}	0.0018
T_{mtp-H}	4.4060
T_{o-H}	0.0001

prime integer, and q is a 160-bit Solinas prime number [29]. The ECC at the 80-bit security level is designed as follows: A point P on a non-singular elliptic curve $E : y^2 = x^3 + cx + d(modp)$ generates an additive group G, whose order is q where $c, d \in Z_q^*$ & p, q are 160-bit prime numbers. Here Table 2 lists some cryptographic operations & their execution time and they are described as follows:

- T_{BP}: Bilinear pairing ET.
- T_{m-BP}: Scale multiplication in relation to the ET of bilinear pairings.
- T_{a-BP}: A bilinear pairing-related point addition procedure.
- T_{m-ECC}: Scale multiplication operation ET for ECC.
- T_{a-ECC}: Point addition operation ET for ECC.
- T_{mtp-H}: ET for map to point hash function operation for bilinear pairing.
- T_{o-H}: ET for one way hash function operation.

We examine the effectiveness of our suggested CL-AS system & further observe its comparison to some schemes in [2], [9], [20], [22], [34], and [36]. Table 3 summarizes the computational expenses for signing a message, validating an individual message, and verifying n messages.

Let us take the experimental data of the scheme given in [9] in which one T_{m-ECC}, one T_{a-ECC} and one T_{o-H} is needed. So the total expense of signing a message is $T_{m-ECC} + T_{a-ECC} + T_{o-H} = 0.442 + 0.0018 + 0.0001 = 0.4439$ ms. Further, we evaluate the expense of verifying a single message which comes out to be 1.3298 ms since it requires three T_{m-ECC}, two T_{a-ECC} and two T_{o-H}. For the aggregate verification, the total computation cost comes out to be (0.8876 + 0.4458n) ms.

Now, we will examine the total computation expense of verifying an individual message in our suggested scheme in which we used five T_{m-ECC}, four T_{a-ECC} and four T_{o-H}. So the complete expense of verifying an individual message comes out to be 2.2176ms, and the total verification cost for verifying n messages comes out to be (1.7734 + 0.4442n)ms. We can also calculate the cost of computation of other schemes in [2], [12], [20], [22], [34], [36], and [38] in the same way as we have done above. Table 3 gives the complete data for the verification cost of a single message and n messages for the above-mentioned schemes. Also, from Table 3, we can see that the amount of producing a digital signature of our suggested technique is greater than Cui et al.'s [9] technique, but we can see that their technique is not secure and can't be utilized for practical purposes.

TABLE 3. Comparison of some related schemes with proposed scheme.

Scheme	Cost of individual verify	Cost of aggregate verify	Pairing	Secure
Cui et al. [9]	$3T_{m-ECC}$ + $2T_{a-ECC}$ + $2T_{o-H}$ = 1.3298 ms	(n+2)T_{m-ECC} + (n+2) T_{a-ECC} + 2nT_{o-H} = (0.8876 + 0.4458n)ms	No	No
Thumbur et al. [34]	$3T_{m-ECC}$ = 1.326ms	(n+2) T_{m-ECC} = (0.442n + 0.884)ms	No	Yes
Ali et al. [2]	$3T_{m-ECC}$ = 1.326ms	(2n+1) T_{m-ECC} = (0.884n + 0.442)ms	No	Yes
Wu et al. [36]	T_{o-H}+ $2T_{m-BP}$ + $3T_{BP}$ = 16.051ms	nT_{o-H} + 2n T_{m-BP} + (2n+3) T_{BP} = (11.84n + 12.633)ms	Yes	Yes
Kamil et al. [20]	$2T_{m-ECC}$ + T_{a-ECC} + T_{o-H} = 0.8859ms	$2T_{m-ECC}$+ nT_{a-ECC}+ nT_{o-H} = (0.884 + 0.0019n)ms	No	Yes
Kumar et al. [22]	$4T_{BP}$ + $3T_{m-BP}$ + $2T_{mtp-H}$ + $3T_{o-H}$ = 30.7833 ms	$4T_{BP}$ + 3nT_{m-BP} + 2nT_{mtp-H} + 3nT_{o-H} = (16.844 + 13.9393n)ms	Yes	Yes
Xiong et al. [38]	$2T_{m-ECC}$ + T_{o-H} = 0.884ms	(n+1)T_{m-ECC} = (0.442n + 0.442)ms	No	Yes
Gong et al. [12]	$4T_{m-ECC}$ + $2T_{a-ECC}$ + $3T_{o-H}$ = 1.7746ms	(2n+1)T_{m-ECC} + 2nT_{a-ECC} + 2nT_{o-H} = (0.4420 + 0.8878n)ms	No	Yes
Proposed Scheme	$5T_{m-ECC}$ + $4T_{a-ECC}$ + $4T_{o-H}$ = 2.2176ms	(n+4) T_{m-ECC} + (n+3) T_{a-ECC} + 4n T_{o-H} = (1.7734 + 0.4442n)ms	No	Yes

Previously, Kamil et al. also mounted the Type II adversary attack over Cui et al.'s scheme and proposed a new scheme. In this paper, we have mounted a universal forgery attack over the same scheme. However, our proposed scheme resists both universal forgery attacks and Type II adversary attacks. Therefore, our scheme is more practical in terms of performance than other proposed schemes.

IX. CONCLUSION

We established in this work that the previously suggested certificateless aggregate signature technique by Cui et al. [9] is not secure. Eventually, we made some relevant changes in the scheme and then introduced a novel, efficient certificateless aggregate signing system based on ECC for VANETs. We used the Discrete Logarithm assumption to establish the security of the proposed approach in the ROM [3]. Furthermore, we evaluated the scheme's security and determined that it met all of the security and privacy standards for Vehicular Adhoc Networks. Immense performance research revealed that our suggested scheme is significantly more efficient in regard to evaluating intricacy than already proposed widely used techniques. Our results indicate that our suggested approach is highly suitable for utilizing in Vehicular Adhoc Networks.

Future Directions: The scheme we have proposed in this paper is secure against various attacks and is more practical,

although the total signing cost is quite larger than the other proposed schemes. We still need to emphasise the fact that how more modifications can be made to get a secure scheme which saves communication as well as computation costs.

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